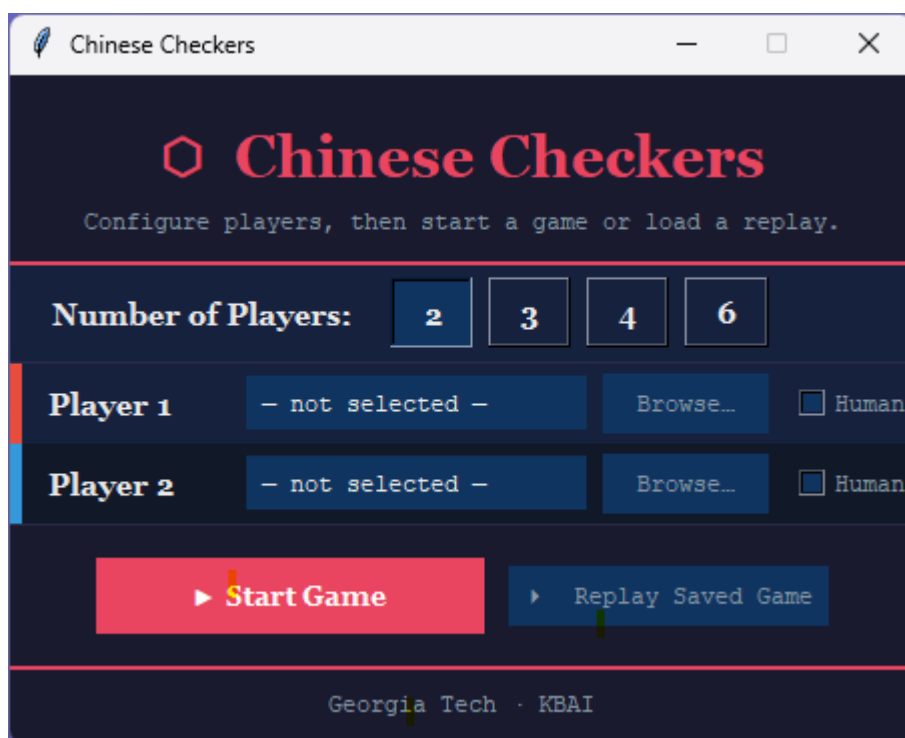


Chinese Checkers

There are a lot of files in the starter pack. You need only concern yourself with modifying the GameAgent.py file. There is only one method in this class that gets called: select_move(...) and you will get a snapshot of the board and a PlayerId (which is your playerId). Normally you wouldn't need it but it was necessary for the GUI playing environment and for playbacks.

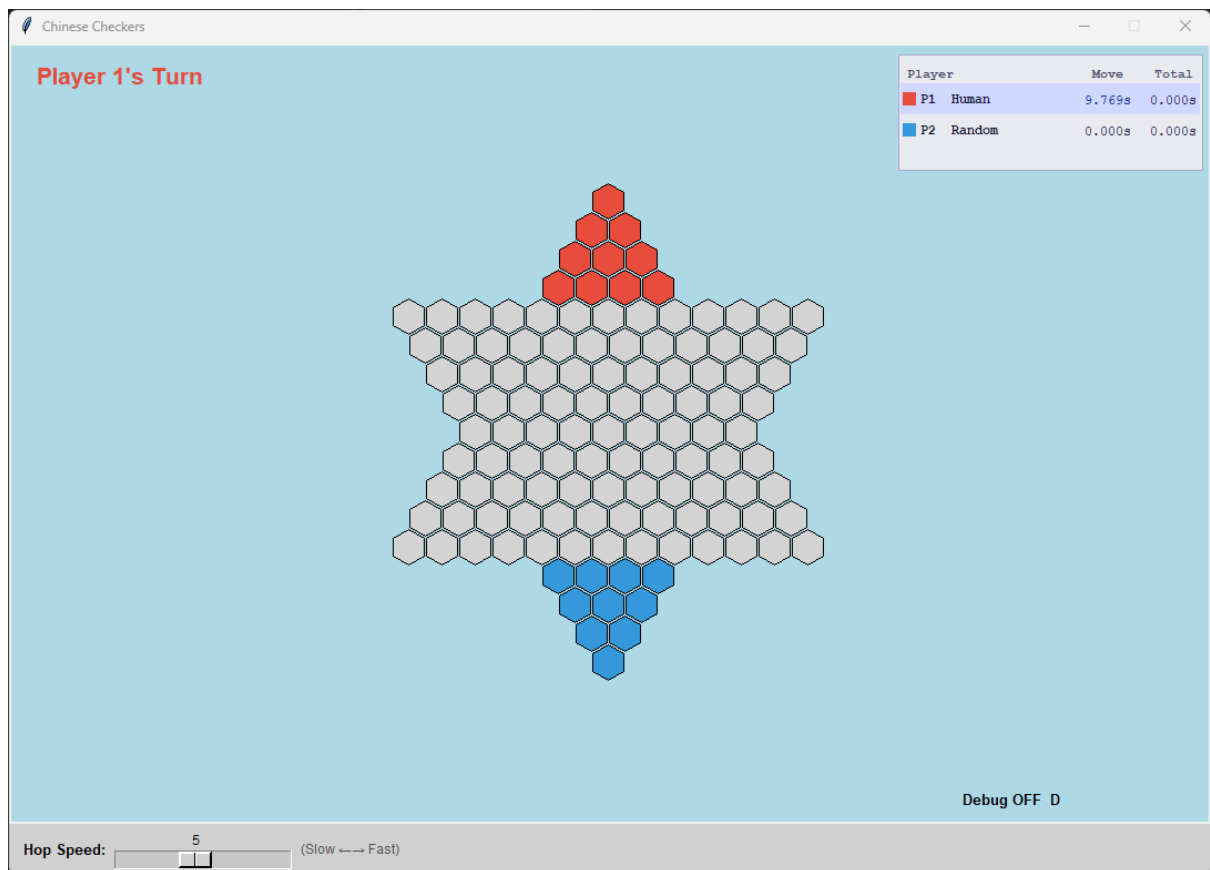
How To Launch A Manual Game:

You will use a file called Launcher.py to kick off the game engine. It's a GUI based program built around TKinter. It should only require python and the tkinter libraries to run. There will be no additional libraries allowed for this project.



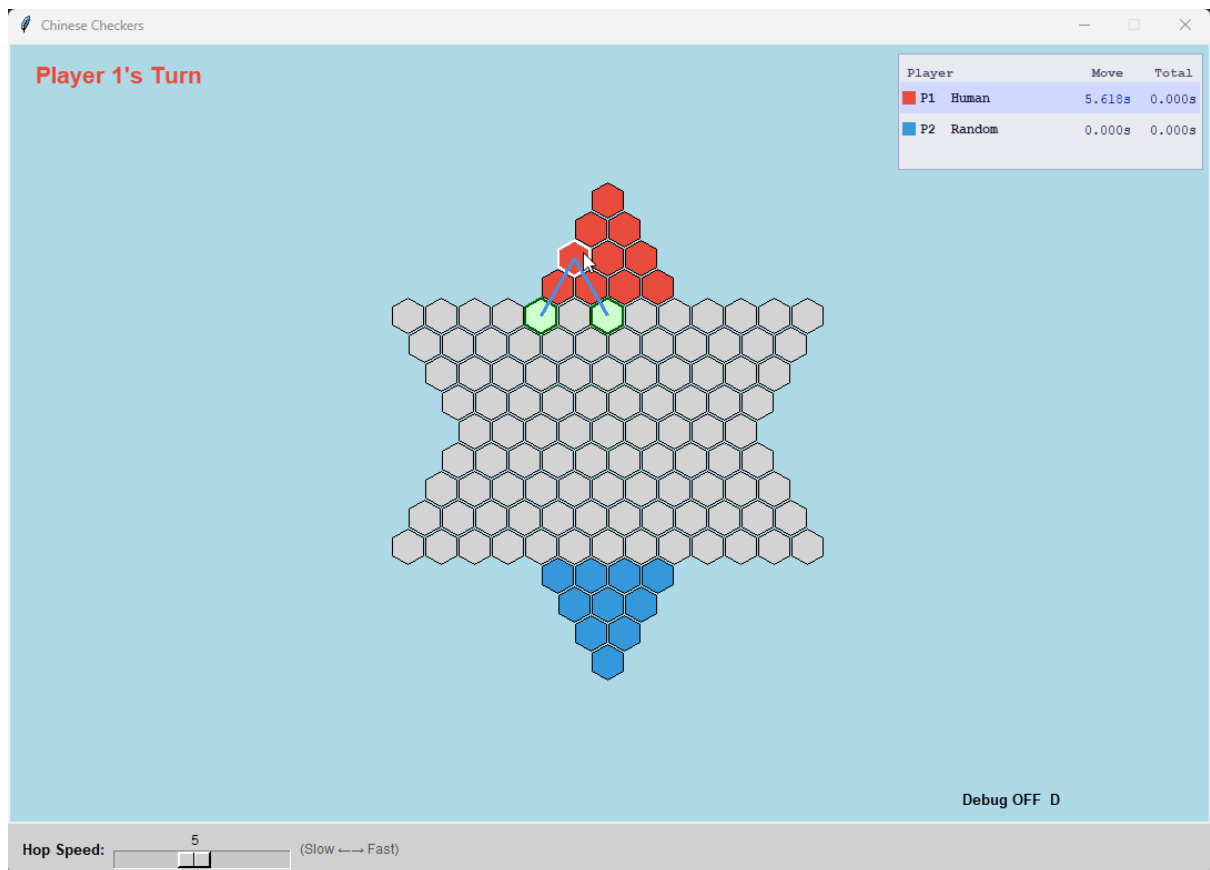
You must select the type of players you want to use for each player. We supplied 3 Agent files that you may use. * GameAgent.py (your agent) * RandomAgent.py (a pure random agent) * HumanAgent.py (a drop in for you to play in the game)

If you are going to play as the Human player you need only click the checkbox (that will load the correct agent). You can also select the number of players you want on the board. The initial offering for this project will be just for 2 players (we may offer additional player versions later in the semester). After selecting the number of players and the player types, pushing Start Game will do just that and you will be presented with the game board:

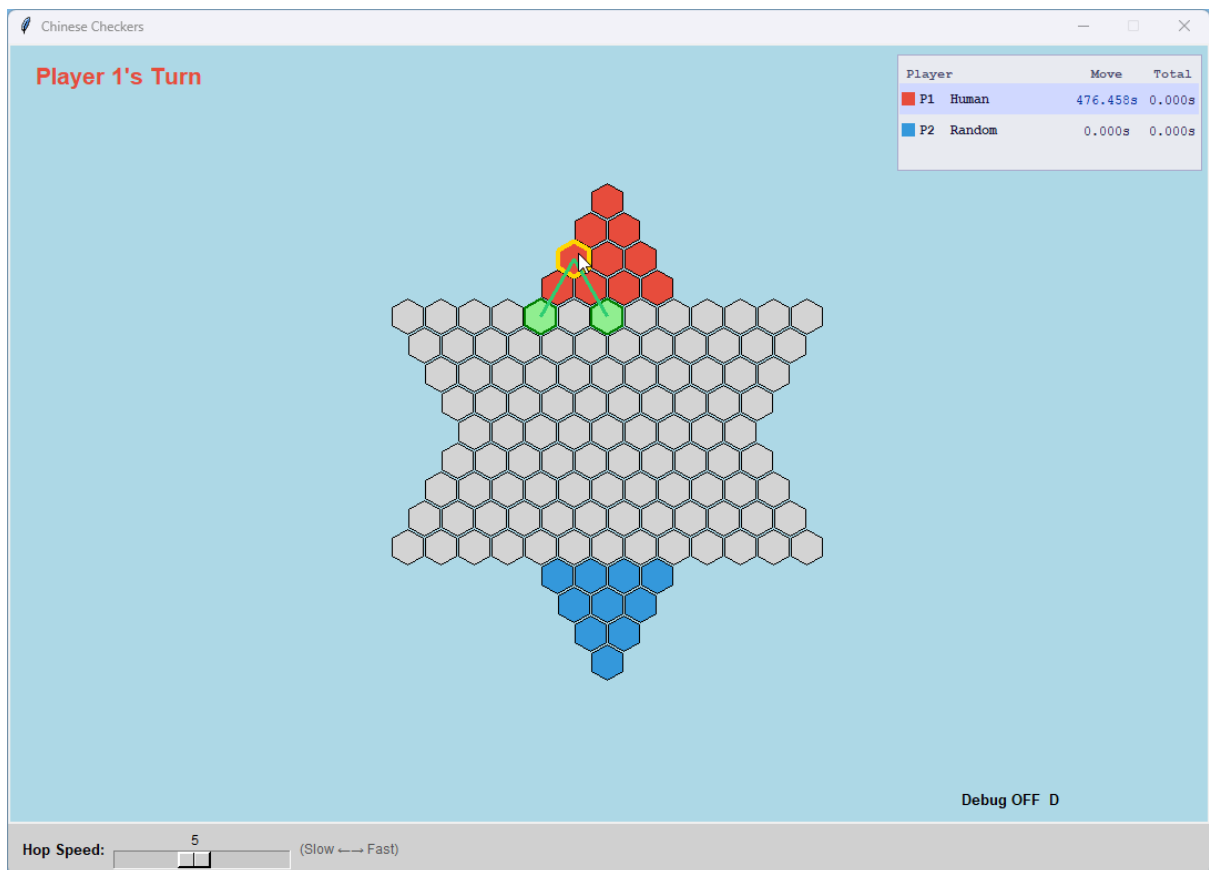


The game board is divided into several sections. The primary board in the center is where you will make your move. You'll notice that the Human player (P1 Red) and the Random Agent (P2 Blue) pieces are automatically placed on the board. In the upper right corner you see each player listed in the game; the amount of time taken for the current move, and a total amount of time taken to play the game (so far) for each of the players. The current players whose turn it is, is also highlighted (and is also displayed in the upper left corner of the window).

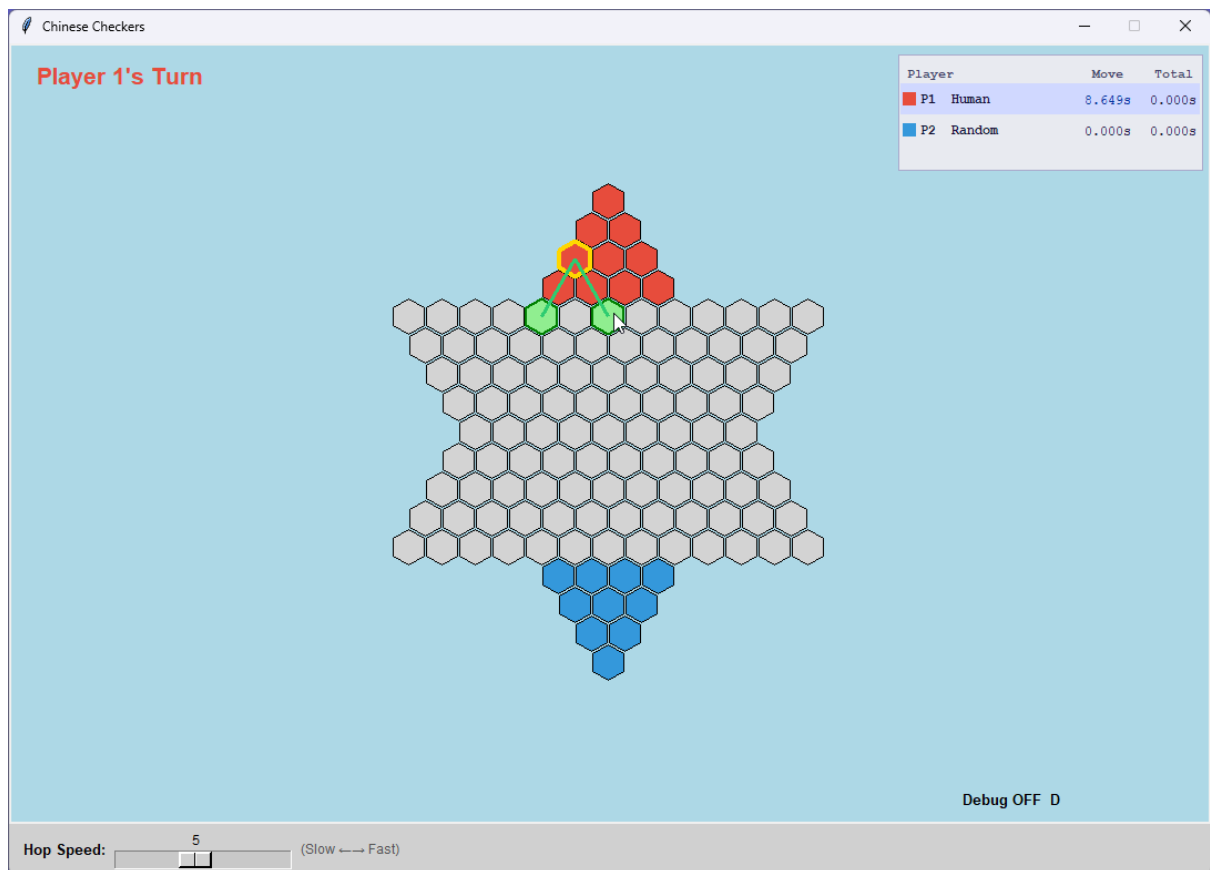
Each player will take turns moving (see the Game Rules for how to play). The human player will be given help in selecting the best moves. Hover the mouse over one of your pieces (Red in this case) and you will be displayed a with all the available moves the agent can make.



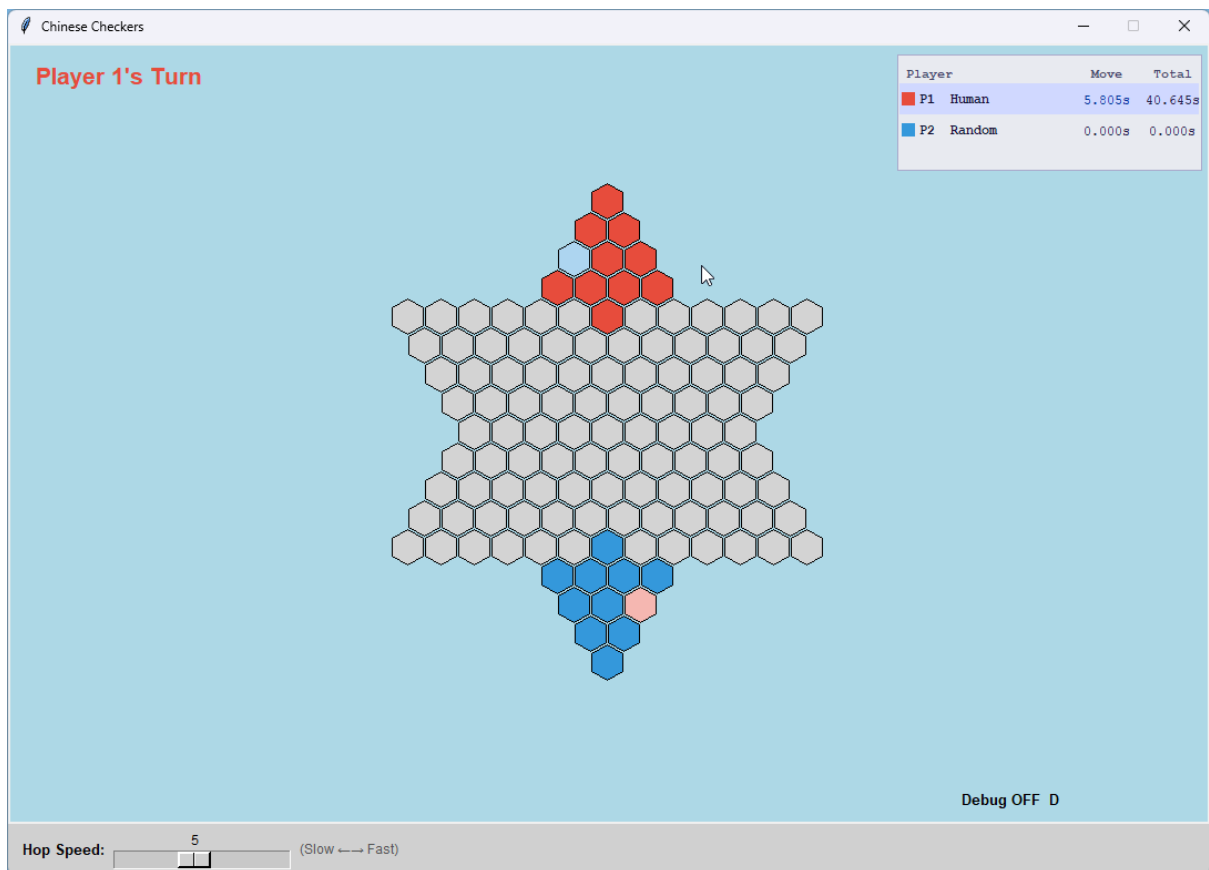
If this is the piece you want to move you then click on it, and it will highlight in Yellow. If you want to select a different piece, you can click on the same piece again , and it will un-highlight, and you are free to choose a different piece.



If this is the piece you want to move then you need to click the ending spot and the piece will be moved.



As soon as your piece has finished its move the Automated Player (Random in this case) turn will commence, and it will immediately select a move and the piece will be moved.



You continue game play until one of the players wins the game or makes an illegal move in which case the game is immediately ended (see the Game Rules for more details). You can even make multiple hops (if there is an open space between 2 pieces ~ yours or an opponents).



You'll notice on the board when either the Random player moved or you moved a piece the space under changes color. It will now show the goal location for the other player (your goal spaces will be Light Red, your opponent's will be Light Blue). This gives you a visual indication of where you may move.

There are a few more features in the game to help you develop your agent. At the bottom of the game board there is a slider that allows you to adjust the timing of the hops. This allows you to speed up or slow down the game as needed.

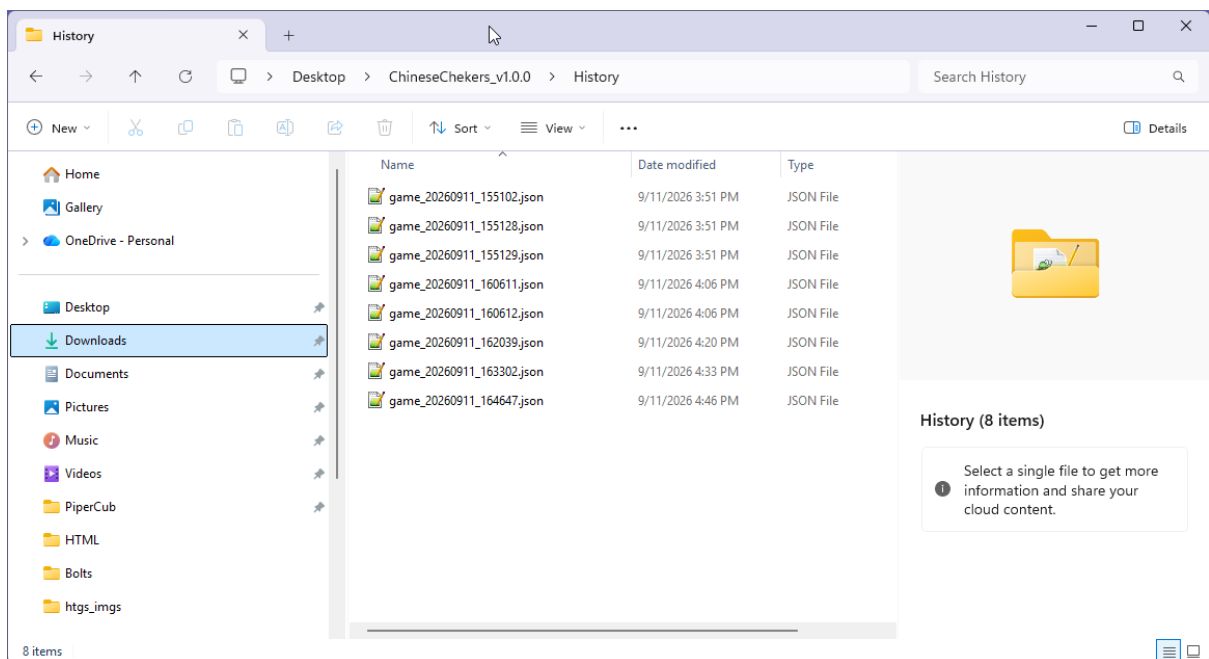
Debug Mode

Another feature is the Debug mode. You activate this by pressing the D key on your keyboard. This will display every cells tuple to aid in debugging your agent if it should make a move that you were not expecting.

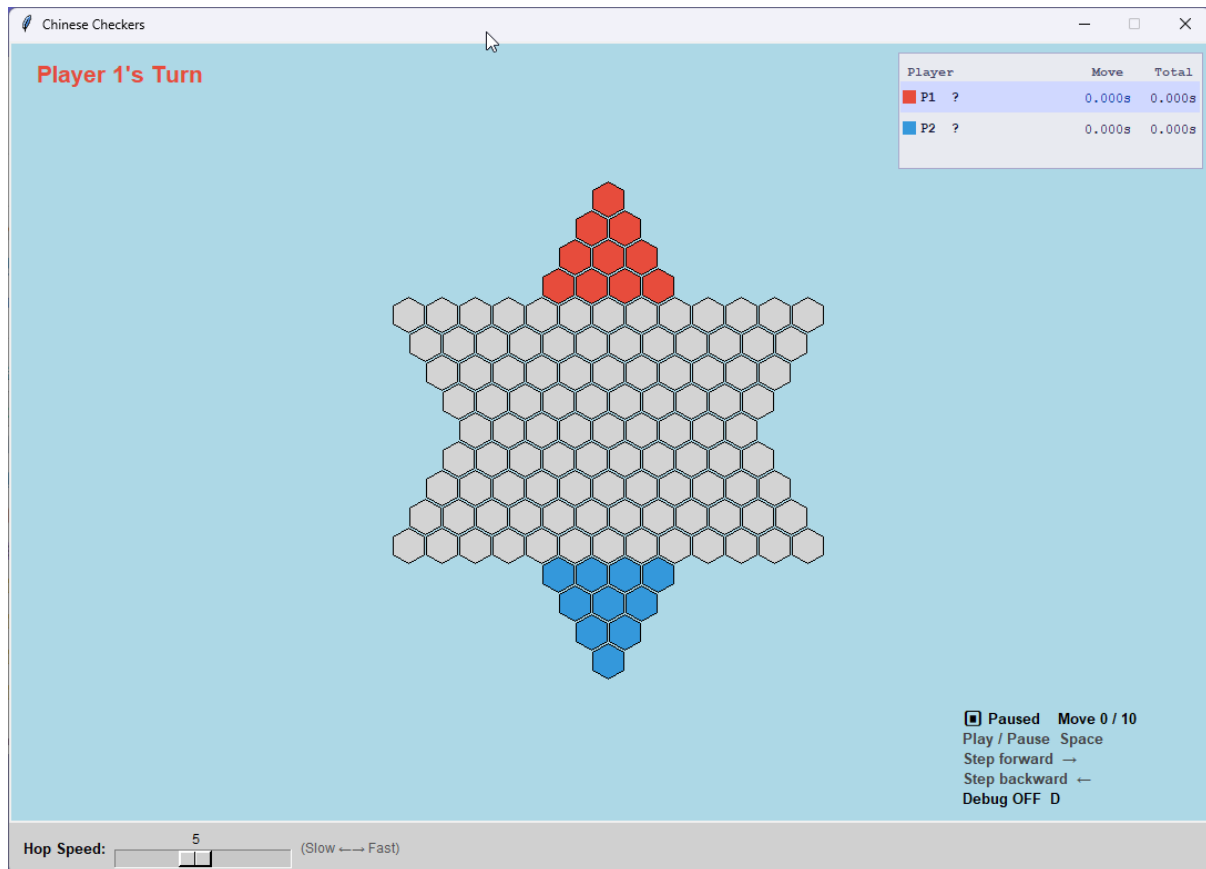


History and Game Playback

The last feature is the Playback. There is a directory in your StarterCode folder called History. Every game (including a partial game) is recorded as a json file here in this directory, and you can play these back.



From the main Launcher window you select the "Replay Saved Game" and you will be presented with a dialog to select a game to be replayed.



To automate the game play back you push the space bar. If you want to manually control the playback you can use the forward and backward arrows on your keyboard to move forward/backward move by move. It also has a move counter so you know where you are at within the game to help you pinpoint moves the agent may be making.

The json history file is pretty straight forward and easy to read. It starts out with some metadata like number of players etc. It then lists all the starting positions for the players. This is followed by the move history, which shows which player made the move then the start and end positions, (some other data that is there for debugging UI issues ~ you won't need it). The elapsed move time is also stored. While you may or may not use these files they will accumulate pretty quickly so you should delete old files when they are no longer needed.